

How to use the Parabolic Microphone to look at the health of our local environment

1

Listening to Sounds



1. Choose a listening spot.
2. Setup the Parabolic Microphone - have a look at the instruction sheet on How to use a Parabolic Microphone.
3. Point the Microphone to listen to the sounds at different locations.
4. Make sure you choose different types of locations to listen to .

2

Give each sound a score

Use the scale bar below to give each sample a score from 1 to 10 (Natural, In-between or Un-natural). Refer to the examples to assist you in completing this task.



Un-natural sounds



Sounds that are entirely made by humans or machines.

Examples: cars moving, machine running, and alarms

In-between sounds



Sounds that are started as a natural sound but was recorded, changed, or used by human.

Examples: animal sounds used in movies, nature sound apps, wind chimes, rain hitting a metal roof

Natural sounds



Sounds that are made by things in nature.

Examples: birds singing, insects buzzing, frogs croaking, and leaves rustling

3

Record your Results

- Record the details of the samples you collected and their score in the Data Collection Sheet.
- Give reasons for each score in the last column.

How to analyse the data and reflect on the results

1

Use your results to calculate the Average Score:



- Add the totals on your data table (in the blue and green boxes)

Sound Type Describe what you think the sound is	Sound Location	How often? Record how long or how many times you hear a sound for	Score Give it a score using the Sound Score bar above	Reason Explain why you gave the score you did.
Bird Chirping, maybe a Rosella	In the tree near basketball courts	Every 10 minutes	10	It came from a bird - possibly a Rosella.
Wind chimes	Front of someone's house	Constant but louder when the wind blew	5	Made from the wind and natural materials (shells) but created by humans.
Bird song recording	Information board near the lake	When button pressed	7	A real bird sound but coming from a speaker
Dog barking	Kindergarten playground	When button pressed	3	Toy dog that makes barking sounds
Total Number of Samples: 4		Total time spent recording data: 30 minutes	Total Score: 25	

- Calculate the Overall rating using the equation (copy from the blue and green boxes):

Average Score = $\frac{\text{Total Score}}{\text{Total Number of Samples}}$

Average Score = $\frac{25}{4}$

= 6.25

2

What does your result tell you? Think back to your original question (from the 'Ask' Stage)

The average score of the items in this environment is 6.25 out of 10. Overall the score for this environment is slightly higher than "in-between", so slightly natural.